7 **(a)** Starting from the definitions of cosh and sinh in terms of exponentials, prove that

$$2\sinh^2 A = \cosh 2A - 1.$$ [3]

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(b) A curve has equation $y = x^2$, for $0 \leqslant x \leqslant \frac{2}{3}$. The area of the surface generated when the curve is rotated through 2π radians about the x-axis is denoted by S.

Use the substitution $x = \frac{1}{2}\sinh u$ to show that $S = \frac{1}{32}\pi\left(\frac{820}{81} - \ln 3\right)$. [9]

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